

ICM-42688-P PIPELINE

Introduction

The ICM-42688-P is a IMU unit which consist of a gyro, accelerometer and a temperature sensor. The device is configurable by I2C, I3C & SPI serial communication, features a 2KB FIFO & 2 programmable interrupts & ultra low power mode. This file will be going over and extracting the needed information to use the ICM-42688-P with a Microcontroller using bare-metal C99.

The device includes industry first 20-bits data format support in FIFO for high data resolution. This FIFO format encapsulates 19-bits of gyroscope data and 18- bits of accelerometer data.

Parameter	ICM-40607 Sensorhub	ICM-42605 Sensorhub	ICM-42686-P Handheld Action	ICM-42688-P HMD & Robotics
GYRO Noise (mdps/rt-Hz)	7	3.8	5.3	2.8
GYRO Offset Temp Stability (mdps/°C)	±30	±20	±10	±5
GYRO Range & Resolution	±2000 dps; 16- bits	±2000 dps; 16- bits	±4000 dps; 16/19- bits	±2000 dps; 16/19- bits
ACCEL Noise (µg/rt-Hz)	110	70	70	AX: 65; AZ: 70
ACCEL Range & Resolution	±16 g; 16-bits	±16 g; 16-bits	±32 g; 16/18-bits	±16 g; 16/18-bits
ODR & Sample Sync	8 kHz; No RTC	8 kHz; No RTC	32 kHz; RTC	32 kHz; RTC

Features

Gyroscope features:

- Digital output (x,y,z) angular rates with programmable scales of ±15.625, ±31.25, ±62.5, ±125, ±250, ±1000, ±2000 deg/sec
- Low Noise (LN) & Low power (LP) modes supported
- Digitally-programmable low-pass filters
- Factory calibrated sensitivity scale factor
- self test

Accelerometer features:

- Digital output (x,y,z) with programmable full scale (±2g, ±4g, ±8g and ±16g)
- Low Noise (LN) & Low power (LP) modes supported
- User-Programmable interrupts
- Wake on motion interrupt

Motion features:

- Pedometer: tracks step count, also issues step detect interrupt
- Tilt detection: issues interrupt when the tilt angle exceeds 35 deg
- Raise to wake/sleep
- Tap detection
- Wake on motion

Additional features:

- External clock input, reduce system level sensitivity error.
- 2k byte fifo buffer enables the applications processor to read in burst
- 20-bits data format support in FIFO for high data resolution
- User programmable digital filters for gyro, Acc and temperature sensor
- 12.5 mHz I3C / 1MHz I2C & 24MHz SPI
- digital output temp sensor
- 20,000 g shock tolerant
- MEMS structure hermetically sealed and bonded at wafer level
- RoHS and Green compliant.

SPI interface

SPI will be the chosen communication protocol between the sensor and the controller for reliability and higher serial data output (SDO). The ICM-42688-P supports 3 & 4 wire SPI for the host interface, the device always operates as slave device during Master-Slave SPI operation.

SPI Operational features:

1. data is delivered MSB first and LSB last (big endian)
2. Data is latched on rising edge of SCLK
3. Data is transitioned on falling edge of SCLK
4. Maximum frequency is 24MHZ
5. SPI reads are completed in 16 or more cycles (2 or more bytes)
6. SPI writes are completed in 16 cycles (2 bytes)
7. R/W first byte contains address & first bit R/W operation / remaining bytes are data
8. Supports single or burst read & writes

RP2350 Example R/W SPI:

```
static inline void cs_low(void) { gpio_put(PIN_CS, 0); }

static inline void cs_high(void) { gpio_put(PIN_CS, 1); }

static void reg_write(uint8_t reg, uint8_t val) {

    uint8_t tx[2] = { (uint8_t)(reg & 0x7F), val };cs_low();
    spi_write_blocking(IMU_SPI, tx, 2);
    cs_high();
}

static void reg_read(uint8_t reg, uint8_t *dst, size_t n) {
    uint8_t hdr = (uint8_t)(reg | 0x80);
    cs_low();
    spi_write_blocking(IMU_SPI, &hdr, 1);
    spi_read_blocking(IMU_SPI, 0x00, dst, n);
    cs_high();
}
```

Writing:

First operation is turning the first byte into a unsigned byte and using a AND operand, it forces the first bit in the reg byte low then the new array tx becomes the full 16 bit write for the ICM. This operation doesn't variable sizes since each write size is constant and specified in the data-sheet.

Reading:

Each read is followed by a single byte write to the IMC, first written byte has its first bit froced high by using a OR operator with 0x80 and the register value, this new value is written to the ICM, the next read operations are the data from the slave device which are in variable sizes. Since each read has different sizes of data the 3rd variable of the function is size.

FIFO Configuration

The IMC-42688-P contains a 2K byte FIFO register accessible via the serial interface. The FIFO configuration register is what determines which data is written into FIFO.

The package structure:

Packet 1	Packet 2	Packet 3	Packet 4
Header (1 byte) ACC data (6 bytes) Temp (1 byte)	header (1 byte) Gyro (6 bytes) Temp (1 byte)	header (1 byte) ACC (6 bytes) Gyro (6 bytes) Temp (1 byte) Time (2 bytes)	Header (1 byte) ACC (6 bytes) Gyro (6 Bytes) Temp (2 bytes) Time (2 bytes) ext (3 bytes)

Package structures are used to transmit different modes, packet 1 transmits as only accelerometer mode, packet 2 is gyro, packet 3 is ACC + gyro + time, packet 3 is same as 23 but with 3 extra bytes. In this case packet 3 will be used since the benefits of packet 4 are negligible in most use cases not involving a well filtered external environment.

Packet 3 content:

Byte	Content
0x00	FIFO Header
0x01	Accel X [15:8]
0x02	Accel X [7:0]
0x03	Accel Y [15:8]
0x04	Accel Y [7:0]
0x05	Accel Z [15:8]
0x06	Accel Z [7:0]
0x07	Gyro X [15:8]
0x08	Gyro X [7:0]
0x09	Gyro Y [15:8]
0x0A	Gyro Y [7:0]
0x0B	Gyro Z [15:8]
0x0C	Gyro Z [7:0]
0x0D	Temperature[7:0]
0x0E	TimeStamp[15:8]
0x0F	TimeStamp[7:0]

FIFO CONFIG1 Registers & setting: (Reg 0x5F : Write 0x47)

BIT	NAME	FUNCTION
4	FIFO_HIRES_EN	0: Default setting; Sensor data have regular resolution 1: Sensor data in FIFO will have extended resolution enabling the 20 Bytes packet with priority on other setting below
3	FIFO_TMST_FSYNC_EN	0: FIFO will only contain ODR timestamp information 1: FIFO can also contain FSYNC time and FSYNC flag for one ODR after an FSYNC event
1	FIFO_GYRO_EN	0: Default setting; Gyroscope data not placed into FIFO 1: Enables gyroscope data packets of 6-bytes to be placed in FIFO
0	FIFO_ACCEL_EN	0: Default setting; Accelerometer data not placed into FIFO 1: Enables accelerometer data packets of 6-bytes to be placed in FIFO

FIFO_HIRES_EN	FIFO_ACCEL_EN	FIFO_GYRO_EN	FIFO_TMST_FSYNC_EN	Header	Packet size
1	X	X	0	8'b_0111_10xx	20 Bytes
1	X	X	1	8'b_0111_11xx	20 Bytes
0	1	1	0	8'b_0110_10xx	16 Bytes
0	1	1	1	8'b_0110_11xx	16 Bytes
0	1	0	X	8'b_0100_00xx	8 Bytes
0	0	1	X	8'b_0010_00xx	8 Bytes
0	0	0 X		No FIFO writes	No FIFO writes

FIFO Configuration

FIFO_CONFIG (Address: 22 (16h) | Serial IF: R/W | Reset value: 0x00 | Clock Domain: SCLK_UI)

BIT	NAME	FUNCTION
7:6	FIFO_MODE	00: Bypass Mode (default) 01: Stream-to-FIFO Mode 10: STOP-on-FULL Mode 11: STOP-on-FULL Mode
5:0	—	Reserved

FIFO_CONFIG1 (Address: 95 (5Fh) | Serial IF: R/W | Reset value: 0x00 | Clock Domain: SCLK_UI)

BIT	NAME	FUNCTION
7	—	Reserved
6	FIFO_RESUME_PARTIAL_RD	0: Partial FIFO read disabled, requires re-reading of the entire FIFO 1: FIFO read can be partial, and resume from last read point
5	FIFO_WM_GT_TH	Trigger FIFO watermark interrupt on every ODR (DMA write) if FIFO_COUNT ≥ FIFO_WM_TH
4	FIFO_HIRES_EN	Enable 3 bytes of extended 20-bits accel, gyro data + 1 byte of extended 16-bit temperature sensor data to be placed into the FIFO
3	FIFO_TMST_FSYNC_EN	Must be set to 1 for all FIFO use cases when FSYNC is used.
2	FIFO_TEMP_EN	Enable temperature sensor packets to go to FIFO
1	FIFO_GYRO_EN	Enable gyroscope packets to go to FIFO
0	FIFO_ACCEL_EN	Enable accelerometer packets to go to FIFO

FIFO_CONFIG2 (Address: 96 (60h) | Serial IF: R/W | Reset value: 0x00 | Clock Domain: SCLK_UI)

BIT	NAME	FUNCTION
7:0	FIFO_WM[7:0]	Lower bits of FIFO watermark. Generate interrupt when the FIFO reaches or exceeds FIFO_WM size in bytes or records according to FIFO_COUNT_REC setting. Interrupt only fires once. This register should be set to non-zero value, before choosing this interrupt source.

FIFO_CONFIG3 (Address: 97 (61h) | Serial IF: R/W | Reset value: 0x00 | Clock Domain: SCLK_UI)

BIT	NAME	FUNCTION
7:4	—	Reserved
3:0	FIFO_WM[11:8]	Upper bits of FIFO watermark. Generate interrupt when the FIFO reaches or exceeds FIFO_WM size in bytes or records according to FIFO_COUNT_REC setting. Interrupt only fires once. This register should be set to non-zero value, before choosing this interrupt source.

FIFO reg / Write Values:

FIFO_CONFIG (0x16 / 0x40) | 01000000

FIFO_CONFIG_1 (0x5F / 0x47) | 01000111

FIFO_CONFIG_2 (0x60 / 0x20) | 00100000

FIFO_CONFIG_3 (0x61 / 0x00) | 00000000

FIFO_CONFIG is used to configure the stream, This application will use Stream to fifo Mode.
FIFO_CONFIG_1 enables Acc[0], gyro[1], temp[2] and FIFO_RESUME_PARTIAL_RD[1] this registers main use case is to enable sensors IF0_RESUME_PARTIAL_RD is used to resume from last read giving a cleaner stream. FIFO_CONFIG_2 is used as a water mark, the watermark is what generates the interrupt for addressing that the fifo is full increasing decreases the workload on SPI but increases the latency. FIFO_CONFIG_3 is used for for watermarks larger than 256 bytes.

INTERRUPT configuration

Which ICM-42688-P settings are best for inertial navigation (4 Technical report)

Of the four parts, the ICM-42688-P is the clear pick for a strapdown INS – it has the lowest gyro noise, the tightest offset temp stability, and the 32 kHz ODR with RTC time-stamping. But the headline specs only get you half the way; the part's actual INS performance is set by how you configure it. Here's what matters, in order of impact on navigation error growth:

- 1. Gyro full-scale – trade measurement range for effective resolution.** The "16/19-bits" resolution is a fixed-point format: full-scale stays 2^{19} counts regardless of FSR, so the scale factor per LSB scales linearly with the range you select. Angle error from quantization is roughly $\sigma_q \approx (\text{FSR} / 2^{19}) \cdot (1/\sqrt{12})$ per sample; at ± 250 dps that's ≈ 0.004 mdps/LSB, vs 0.03 mdps/LSB at ± 2000 dps. For platform stabilization or slow-moving navigation, use **± 250 or ± 500 dps** (GYRO_CONFIG0, register 0x4F). Only go to ± 2000 dps if you have genuine body rates above a few hundred dps – high vibration environments or fast attitude maneuvers – because 2.8 mdps/ $\sqrt{\text{Hz}}$ noise density matters more than quantization until FSR gets large.
- 2. Gyro ODR and anti-alias filtering.** The 32 kHz ODR (GYRO_CONFIG0, 32 kHz mode) is valuable for two reasons: it averages the integrated rate over shorter windows, and more importantly it lets you push the **gyro anti-alias filter (AAF, bank 1, registers 0x0C–0x0D)** low enough to reject structural vibration before decimation. For a vehicle or UAV, set AAF corner $\sim$ half your decimated rate. Note that the accel is limited to 32 kHz as well, but the usable signal bandwidth tops out around a few kHz – don't confuse ODR with bandwidth.
- 3. Offset stability vs. bias calibration.** The ± 5 mdps/ $^{\circ}\text{C}$ spec is the table's best by a factor of 2, but understand what it is and isn't: it's the *slope* of zero-rate offset vs. temperature. In an INS, attitude error from an uncorrected gyro bias b grows linearly: $\Delta\theta(t) \approx b \cdot t$ (plus the noise-driven random-walk term, $\Delta\theta \sim N \cdot \sqrt{(2T)}$ with $N = 2.8$ mdps/ $\sqrt{\text{Hz}} \approx 0.162^{\circ}/\sqrt{\text{hr}}$ ARW-class performance). A 5 mdps/ $^{\circ}\text{C}$ slope means a 20°C swing uncalibrated injects 100 mdps $\approx 0.34^{\circ}/\text{s}$ of error – that dominates everything. So: temperature-compensate with an offline calibration (store bias as a function of temperature in firmware), or keep the IMU thermally controlled. With calibration you reduce the effective bias to the hysteresis level, which is what separates good dead reckoning from drift within minutes.
- 4. Accel range and noise anisotropy.** Use **± 16 g** (ACCEL_CONFIG0) – you don't need ± 32 g on this part, and staying at ± 16 g maximizes your effective resolution. Note the spec sheet's quirk: X-axis noise density is 65 $\mu\text{g}/\sqrt{\text{Hz}}$ vs 70 for Y and Z. For a Z-up or X-forward navigation frame, the difference is marginal, but it's worth mounting the part so the lowest-noise axis (X) aligns with the most velocity-critical direction. The accel bias stability isn't in this table – check the full datasheet's bias repeatability specs, because accel bias integrates directly into velocity error, $\Delta v \approx b_a \cdot t$, and tilt misalignment couples gravity (~ 2.4 mm/s of pseudo-velocity error per milliradian of tilt, per g).
- 5. RTC and time alignment.** This is the unsung advantage of the -P variants: the 32 kHz sample clock synchronized to the internal RTC (via the TMST registers) gives you sub-microsecond-consistent time stamps on every FIFO sample. For INS/EKF fusion with GNSS, odometry, or vision, timestamp error Δt on a rotating body creates apparent attitude error $\omega \cdot \Delta t$ – at $500^{\circ}/\text{s}$, 1 ms of time error is already 0.5° . Read TMST_S0 (0x62) and TMST_S1 (0x63) in user bank 0 to get the RTC-synced FIFO timestamps.

Practical register setup for navigation (bank 0):

- PWR_MGMT0: accel + gyro on in LN (low-noise) mode – avoid LP (low-power) mode; it uses duty-cycled sensing with worse bias
- GYRO_CONFIG0: ± 250 – 500 dps, 32 kHz (or 8 kHz if power budget forces it; 32 kHz if you want the AAF margin)
- ACCEL_CONFIG0: ± 16 g, 32 kHz
- GYRO_CONFIG1: UI filter order 3rd (lowest group delay for real-time fusion)
- Bank_1, GYRO_X/Y/Z_AAF + GYRO_CONFIG_STATIC6–9: AAF corners matched to your signal band
- FIFO_CONFIG / TMST_S0–S1: enable RTC time stamps and stream 20-byte packets (accel, gyro, temp, time)

One caveat from the datasheet itself: you can only change ODR, FSR, and operating mode while the sensors are running – every other register write requires both accel and gyro off first (register 12.9), so build your init sequence to set all filters and configs before enabling sensors, or accept a re-init cycle when changing them.

The one thing no register setting fixes: the gyro's long-term bias stability (the "instability" term in an Allan deviation) isn't specified in this table, so if you're doing minutes-long dead reckoning without GNSS aiding, plan on periodic zero-velocity or magnetometer updates regardless of how well you calibrate temperature.